

# Smartwatch-based detection of cardiac arrhythmias: Beyond the differentiation between sinus rhythm and atrial fibrillation



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Within the span of a few years, watches have functionally morphed from objects that tell time to wearable minicomputers that allow real-time recording of electrocardiograms (ECGs). Considerable information can be deduced from these single lead tracings, and it is now not uncommon to see patients in whom diagnostic tracings of clinically relevant but elusive arrhythmias are captured using a smartwatch. Empowering individuals to record their own ECG tracings in scenarios such as palpitations, syncope, and for risk stratification of sudden death intuitively has considerable potential, but its value remains to be robustly demonstrated. The main objective of this review is to describe the information that can be ob-

tained from smartwatch-based single-lead ECG recordings beyond simply differentiating between sinus rhythm and atrial fibrillation. We also review the strengths and limitations of using these devices in clinical settings and offer potential solutions to address the latter.

**KEYWORDS** Atrial fibrillation; Electrocardiogram; Smartwatch; Tachycardia; Ventricular tachycardia

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## A. Introduction

The rapid and widespread introduction of smartwatches in health care presents both unique opportunities and challenges. Within the span of a few years, watches have functionally morphed from objects that tell time to de facto wearable minicomputers. Consumers from the general public now have direct access to devices and applications that offer real-time recordings of electrocardiogram (ECG) tracings. The tracings are recorded from 2 electrodes, one located on the back of the watch in contact with the skin and the other positioned on the watch casing (usually the digital crown). The tracing is automatically analyzed by an algorithm included in a proprietary application that measures the heart rate, analyzes the regularity of the rhythm, and searches for evidence of atrial fibrillation (AF). The tracings obtained by these devices differ in important ways from those of standard 12-lead ECGs that are typically performed in clinical settings as only 1 lead (lead I) is recorded. Accordingly, smartwatch tracings are not as comprehensive as 12-lead

ECGs and should not be considered substitutes for them. However, a considerable amount of information, especially relating to the wearer's heart rate, can be deduced from this single lead.

Interest in these types of devices and in their capabilities is growing rapidly.<sup>1,2</sup> It is now not uncommon to see patients in whom diagnostic tracings of clinically relevant arrhythmias proved elusive until a smartwatch was used. Given that cardiac arrhythmias are often difficult to capture using standard investigations, the potential benefits of empowering individuals to record their own ECG tracings in real-time in various clinical scenarios are considerable but remain to be robustly demonstrated.

The main objective of this review is to describe the information that can be obtained from an ECG recording with a single lead beyond the differentiation between sinus rhythm and AF. We also review the strengths and limitations of using these devices in clinical settings and propose solutions to maximize their value.

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## B. Technical considerations

Although there exist important technical differences among available smartwatches and their ECG recordings, they share common features. Three points should be considered when using these devices to diagnose or monitor cardiac arrhythmias:

1. Unlike ambulatory Holter monitoring and implantable loop recorders, smartwatches do not continuously record ECGs. The user must initiate a recording and actively hold the contralateral index finger on the smartwatch. This is a major limitation for their use in patients with recurrent syncope. Initiating an ECG recording requires a few seconds (opening of the related application directly on the watch or on a paired smartphone depending on the model); then the user must hold their finger steadily on the crown for at least 30 seconds. This delay renders existing smartwatch technologies poorly suited for the diagnosis of brief arrhythmias as well (ie, lasting less than a minute). The value of using smartwatch ECG tracings for diagnosing asymptomatic arrhythmias such as AF is also questionable unless their arrhythmic burden is considerable as the patient must proactively initiate recordings without being prompted by symptoms. In Table 1, characteristics of smartwatch ECGs are compared with alternative systems. An important difference between ECG smartwatches and existing ECG systems is that smartwatches are not integrated in current practice, are not reimbursed, and may cost the patient between \$150 (Withings) to more than \$1000 (Apple Watch with iPhone). In the absence of it being covered by health care systems, this cost poses an obstacle to its democratization. The detailed comparison between existing smartwatches is quickly outdated and beyond the scope of the present review, but there exist important differences beyond the price that may have an impact on accessibility. For example, most smartwatches (ie, Apple and Samsung) require charging of the battery during the night, which may hinder the possibility to register an ECG during symptoms. The battery life of other ECG smartwatches may exceed multiple days (Fitbit Sense) or even months (Withings).

2. The growing popularity of smartwatches renders it increasingly likely that false-positive diagnoses of arrhythmias will occur, particularly in users with a low prevalence of cardiovascular disease. To minimize the risk of false positives, optimal recordings should be sought whenever possible, including taking steps to minimize baseline wander and artifacts. This risk is highest when ECG tracings are recorded during strenuous activity because of the combination of limb movement and perspiration. Undulations of the ECG baseline most often occur when there is poor contact between the skin and the electrode or during limb movement, and sweat can affect the quality of the recording, all of which can be problematic during vigorous exercise. For optimal tracing quality, the skin and the watch must be clean and dry. The strap must also be properly adjusted to ensure stable and permanent contact between the back of the watch and the skin. Most importantly, the wrist should remain static for the duration of the recording.
3. Wrist-worn devices are best suited for measuring ECG lead I, which is not the ideal lead for detecting P waves, thus potentially affecting the interpreter's ability to diagnose the atrial rhythm. Repeating the recording or placing the watch on an alternative position, such as the ankle or the precordium, can facilitate the detection of P waves.

## C. Accuracy of smartwatch for the detection of AF

AF is the most common sustained rhythm disorder encountered in clinical practice and is an increasingly important public health problem, with nearly one-third of all strokes being attributed to this arrhythmia.<sup>3</sup> Wearables have been proposed as screening tools to detect AF both in high-risk groups and in the general population with the goal of initiating anticoagulants to prevent strokes.<sup>4</sup> This technology could similarly be used for patients with known AF to assess the effectiveness of treatments and to plan interventions.<sup>5</sup>

**Table 1** Overview of similarities and differences between out-of-hospital ECG recording systems

| Variable                       | Smartwatch ECG | Ambulatory ECG | Short-term Holter recording | Long-term Holter recording | Implantable loop recorder |
|--------------------------------|----------------|----------------|-----------------------------|----------------------------|---------------------------|
| Palpitations <1 min            | —              | —              | —/+                         | ++                         | +++                       |
| Palpitations >1 min            | +++            | ++             | —/+                         | +                          | +++                       |
| Syncope                        | —              | —              | —/+                         | ++                         | +++                       |
| Asymptomatic AF                | +*             | +              | —/+                         | ++                         | +++                       |
| Patient participation required | +++            | +++            | +                           | ++                         | —                         |
| Costs                          | \$ - \$\$\$    | \$             | \$                          | \$                         | \$\$\$\$                  |

AF = atrial fibrillation; ECG = electrocardiographic.

— means less suitable while +++ means very suitable. \$ means low cost while \$\$\$\$ means very high cost.

\*Efficacy of asymptomatic AF detection is higher when combined with a photoplethysmography alert.

### C.1. Semiautomated diagnosis of AF

After recording suspected AF with a smartwatch, the user may send the tracing to a health care professional for analysis. The ability of physicians to correctly identify AF on smartwatch ECG tracings was first investigated using Kardia Band technology (AliveCor, Mountain View, CA) combined with an Apple smartwatch application (Apple, Cupertino, CA) in 100 patients undergoing cardioversion.<sup>6</sup> The accuracy for this arrhythmia was high with 93% sensitivity and 97% specificity. The authors concluded that when supported by physician review, the system can reliably differentiate AF from sinus rhythm. We reported similar diagnostic accuracy using Apple Watch ECGs.<sup>7</sup> Apple Watch ECGs recorded during AF are generally of sufficient quality to make appropriate diagnoses when they are interpreted by a cardiologist. It is important, however, to stress the need to record a tracing of sufficient quality to avoid false positives, avoiding baseline artifacts that could simulate features of AF, especially in the context of subtle P waves. The option of recording an ECG with a smartwatch and then sending it to a cardiologist could be appealing, especially to patients with symptomatic episodes that are long enough to be recorded (more than a few minutes) but short or infrequent enough to elude capture using typical diagnostic studies.

### C.2. Fully automated diagnosis of AF

Currently available smartwatches have in part been marketed for AF detection. Manufacturers have integrated proprietary algorithms on the basis of ventricular rhythm irregularity to automate the diagnosis of AF. Examples of smartwatch ECGs recorded in the same patient with AF using the Apple Watch, Withings Move ECG, and Samsung Galaxy watches are shown in [Figure 1](#).

The purpose of these algorithms is solely to differentiate between AF and sinus rhythm. They do not identify other arrhythmias (eg, atrial flutter, junctional tachycardias, and ventricular tachycardias). Rhythms are classified as sinus rhythm, AF, or poor recording/inconclusive. The ability of the ECG app to classify a tracing as AF or sinus rhythm

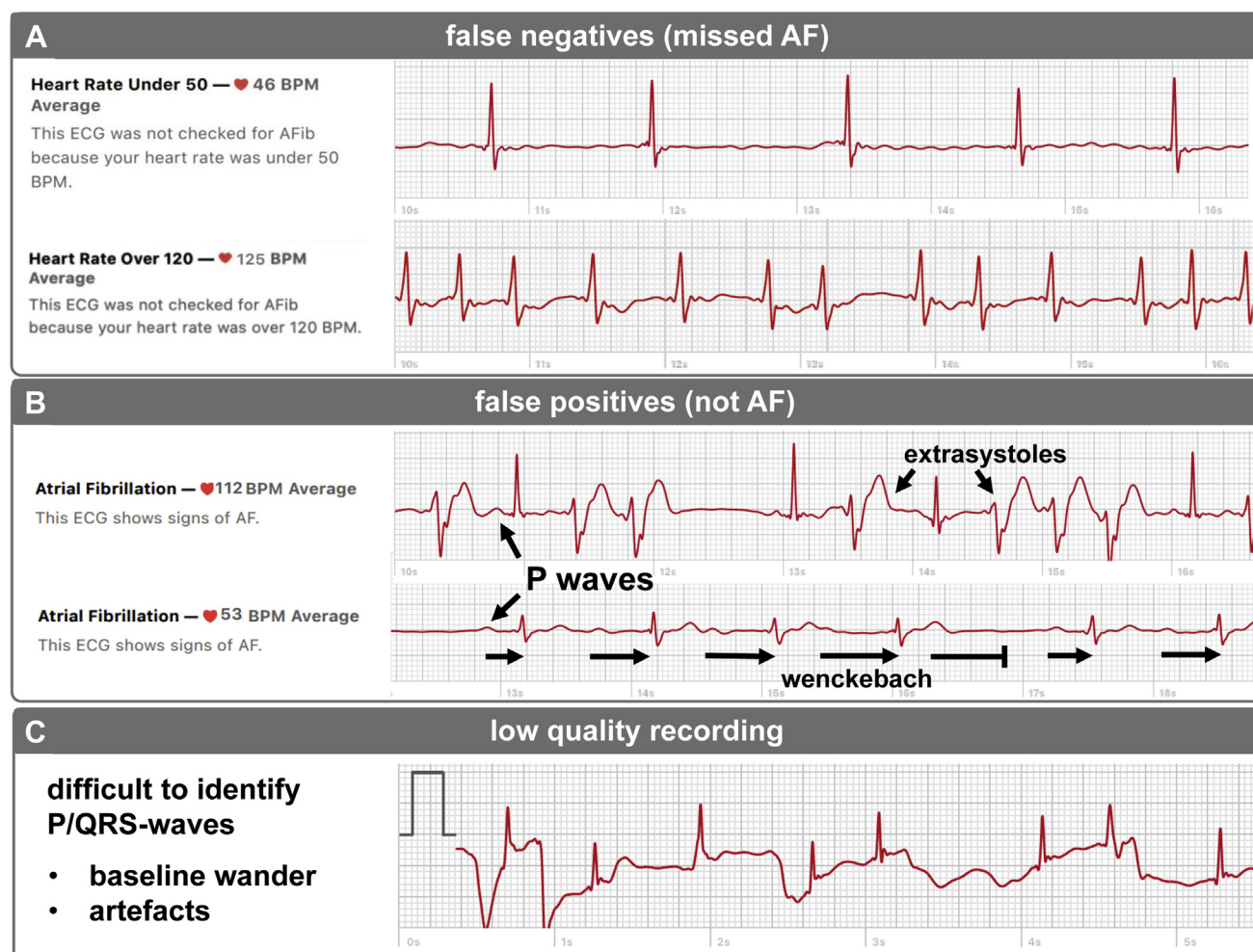
was initially validated in an Apple-sponsored multicenter study of approximately 600 individuals in a medical environment. The ECG app correctly classified the recording as sinus rhythm in 91% and as AF in 85%, whereas 12% of the tracings were inconclusive. It is likely that in real-world settings, the proportion of unclassifiable rhythms would be higher, with a greater frequency of tracings classified as inconclusive. Kardia Band technology's automated interpretation algorithm was similarly reported to have high sensitivity (93%) and specificity (84%) for AF.<sup>6</sup> False negatives (a diagnosis of sinus or inconclusive rhythm when the patient is in AF) can falsely reassure the patient and lead to diagnostic and therapeutic delay ([Figure 2A](#)). Conversely, false positives (a diagnosis of AF when the patient is in sinus rhythm) can lead to unnecessary examinations or initiation of inappropriate treatment ([Figure 2B](#)). Inadequate or variable skin-to-electrode contact may induce artifacts and baseline drift that may render the tracing difficult to interpret ([Figure 2B](#)). In order to protect the reader and the algorithm against false positives or false negatives, the smartwatch will automatically interrupt and reject the recording and launch a new recording in case of excessive noise during the 30-second recording.

The sensitivity and specificity for AF may differ between the available smartwatches, but this has not yet been adequately investigated. In a recent update to the Apple Watch algorithms, the upper limit for automatic AF detection has been increased from 120 to 150 beats/min. Such differences could mean that certain smartwatches are better suited for certain clinical situations than others. There currently exists a mismatch between the population most targeted by smartwatch marketing, namely, a rather young population attracted to new technologies but generally in good health, and the population that would likely most benefit from its ECG recording feature, namely, older individuals who may be more reluctant to adopt this technology but who are at a considerably higher risk of cardiovascular disease. In a recent cohort analysis, it was observed that individuals in whom AF was detected who used wearables were 6 years younger, healthier, and socioeconomically better off than those who did not use wearables.<sup>8</sup> It is plausible that generational



**Figure 1** Examples of electrocardiographic (ECG) tracings diagnosed as atrial fibrillation by the Apple Watch (left), Samsung Galaxy watch (center), and Withings Move ECG watch (right). AF = atrial fibrillation.





**Figure 2** False-negative and false-positive diagnoses. **A:** Two examples of missed atrial fibrillation (AF) diagnoses by the smartwatch because of a ventricular rate that is too slow (*top*) or too fast (*bottom*). **B:** Two examples of irregular rhythms (extrasystoles and second-degree atrioventricular block type I) erroneously diagnosed as AF. **C:** Example of a low quality recording with baseline wander and artifacts, which makes it difficult to identify P/QRS waves. ECG = electrocardiogram.

barriers of using smartwatches by the elderly may persist into the following decade but decline thereafter. As the pretest probability of AF is very low in most smartwatch owners, smartwatches with a high specificity for AF should be favored to limit false-positive results and to minimize patient anxiety and burden to health care systems. Relying on device-based fully automated diagnoses of AF to guide the initiation of oral anticoagulation for stroke prophylaxis cannot be recommended; expert ECG analysis remains indispensable at this time. Conversely, in patients with a known history of AF and who use the smartwatch to monitor the effectiveness of treatments such as AF ablation or antiarrhythmic medications, the sensitivity for AF may be more valuable.

### C.3. Diagnosis of asymptomatic AF

Diagnosing asymptomatic AF can play an important role in reducing the risk of stroke in the general population and properly treating patients with cryptogenic strokes. However, the lack of symptoms can render the above-mentioned

technologies ill-suited unless the arrhythmic burden is high. Several smartwatches offer passive and continuous consumers heart rate monitoring with photoplethysmography (PPG), which relies on infrared light-emitting diode optical sensors to monitor changes in the blood volume passing through the wrist. Each pulse signal captured by PPG can be interpreted as an R wave on the ECG and the peak-to-peak interval between pulsations can be interpreted as the cardiac R-R interval. Different manufacturers have proposed proprietary algorithms on the basis of pulse irregularity to detect AF. The Apple Heart Study relied on such optical sensors in Apple Watch Series 1 through 3 to detect an irregular pulse and a proprietary algorithm to trigger confirmatory testing for AF. Among the nearly 420,000 individuals 22 years and older in the study, 0.52% received notifications of an irregular pulse.<sup>9</sup> Of these, 450 participants underwent ECG patch testing for a mean of 6.3 days, with 34% ultimately found to have AF. The positive predictive value for the irregular pulse notification was 84% on the basis of participants who received this notification while wearing an

ECG patch. Although this study was not designed to determine sensitivity or specificity, which would have required continuous ECG monitoring of all participants, available data suggest that both may be poor. The sensitivity may be relatively low for various reasons: PPG was not recorded during physical activity because of significant noise artifact with arm movement or at night when the device had to be recharged, nearly 4% of PPG tachograms were of insufficient quality for interpretation, and the algorithm is not intended to detect short episodes of AF (although the optimal AF duration threshold to initiate anticoagulation in this context remains unclear). Consumers alerted with irregular heart rhythm notifications by selected smartwatches are requested to record an ECG to confirm the presence of AF. While this feature does not affect its sensitivity for AF detection, it may substantially improve its specificity. This was recently shown in 204 participants using the Samsung Galaxy Active 2 smartwatch. If PPG detected an irregular rhythm over a 5-minute window, a notification was sent to the participant to register an ECG using the smartwatch. This resulted in a near-perfect specificity of 99.3% (validated using simultaneous recordings from a ECG patch).<sup>10</sup> Overall, although population-level screening for AF using smartwatch technology is potentially feasible, important limitations such as its limited sensitivity remain barriers. Furthermore, as with any medical device capable of detecting AF, the question of what AF duration threshold should prompt the introduction of anticoagulant therapy remains unanswered.

#### D. Organized atrial arrhythmias

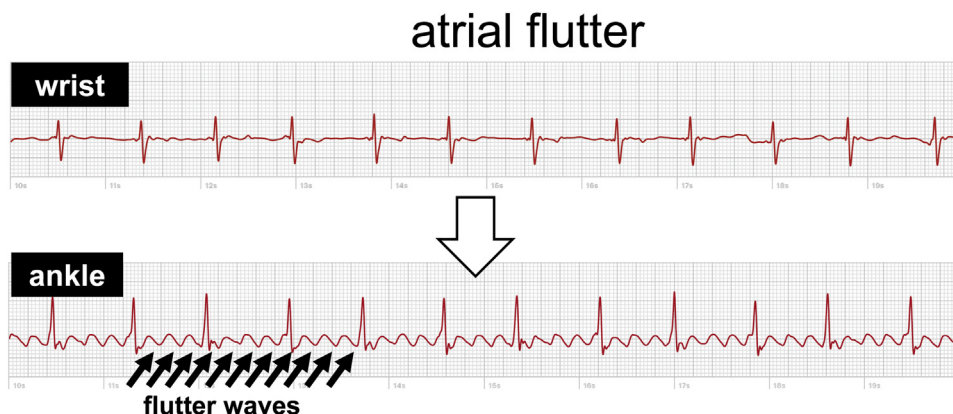
Although its incidence is approximately 10 times lower than that of AF, typical atrial flutter remains relatively common.<sup>11</sup> It has a characteristic appearance on the 12-lead ECG of stable flutter waves (“sawtooth” appearance) in the inferior leads. When the smartwatch is positioned on the left wrist, the diagnosis of typical atrial flutter can be difficult to make. There is usually very low voltage atrial activity in lead I, as it is largely oriented perpendicular to the activation

vector. Therefore, atrial flutter with variable atrioventricular conduction can be mistaken for AF. The distinction is important, particularly if considering ablation therapy, as the former would be limited to the cavotricuspid isthmus whereas the latter involves transseptal access and pulmonary vein isolation. Making the diagnosis of typical atrial flutter with a smartwatch usually requires that an inferior lead equivalent tracing be recorded to confirm the characteristic flutter waves.<sup>12</sup> Figure 3 shows an example of an atrial flutter where atrial depolarization is unmasked when moving the smartwatch from the wrist to the ankle.

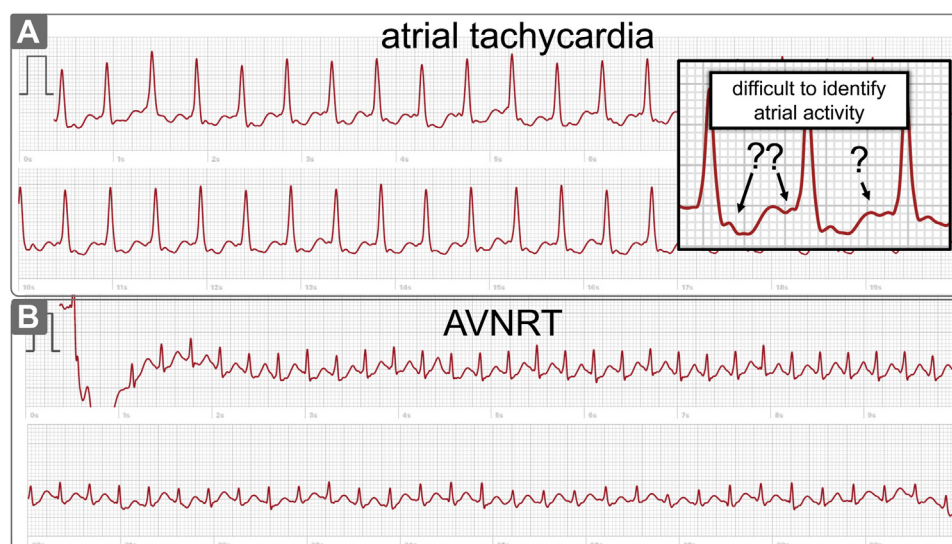
Atrial tachycardia generally refers to an organized rapid atrial rhythm other than cavotricuspid isthmus-dependent flutter and is often seen in the context of previous atrial ablation procedures. Atrial tachycardia is reflected on the ECG by fast, ectopic, usually regular and monomorphic atrial activity that may be difficult to appreciate, with usually regular QRS complexes (although variable atrioventricular conduction can render the ventricular rhythm irregular). The distinction between AF and atrial tachycardia is also important if considering an ablation strategy. Figure 4A shows an example of an atrial tachycardia where atrial activity is difficult to discern because of the single lead and frequent and broad QRS complexes. This issue may be overcome by obtaining additional leads.

#### E. Atrioventricular (nodal) reentry tachycardia: The ideal indication?

Atrioventricular (nodal) reentry tachycardias can occur at all ages but generally manifest in young patients without structural heart disease, which is also the population most likely to wear a smartwatch. Patients usually describe a history of recurrent palpitations of varying duration that resolve spontaneously or with vagal maneuvers. In the vast majority of cases, the arrhythmia is hemodynamically well tolerated, which enables the recording of an ECG with a smartwatch (Figure 4B).<sup>13</sup> Atrioventricular (nodal) reentry tachycardias are regular, usually narrow complex tachycardias, with a 1:1 relationship between P waves and QRS complexes. Atrial activity can be difficult to discern on smartwatch tracings,



**Figure 3** Typical atrial flutter. **Top tracing:** A smartwatch electrocardiogram (ECG) with the watch placed on the wrist with poorly discernible atrial activity. **Bottom tracing:** Smartwatch ECG with the watch positioned on the left ankle (lead II equivalent), showing the classic sawtooth appearance of typical atrial flutter at an atrial rate of 270 beats/min that is regular and monomorphic (arrows).



**Figure 4** **A:** Atrial tachycardia. This tracing shows a regular tachycardia of 140 beats/min. The 12-lead surface electrocardiogram (ECG) confirmed the diagnosis of atrial tachycardia (atrial rate of 280 beats/min, ventricular rate of 140 beats/min, ie, 2:1 atrioventricular conduction). ?: possible atrial activity but impossible to confirm. **B:** AVNRT. The tracing shows a regular narrow complex tachycardia at 186 bpm. The atrial activity is once again difficult to confidently discern, leaving various forms of supraventricular tachycardia possible, including atrial tachycardia, AVRT, and AVNRT.

particularly as it can be hidden within QRS complexes. While the diagnosis of a narrow complex tachycardia is generally easily made with a smartwatch, evaluating the RP interval can prove challenging. To confirm the mechanism of the arrhythmia, an electrophysiology study is often required, as is the case with 12-lead ECGs.

## F. Wide complex tachycardias and the risk of false positives

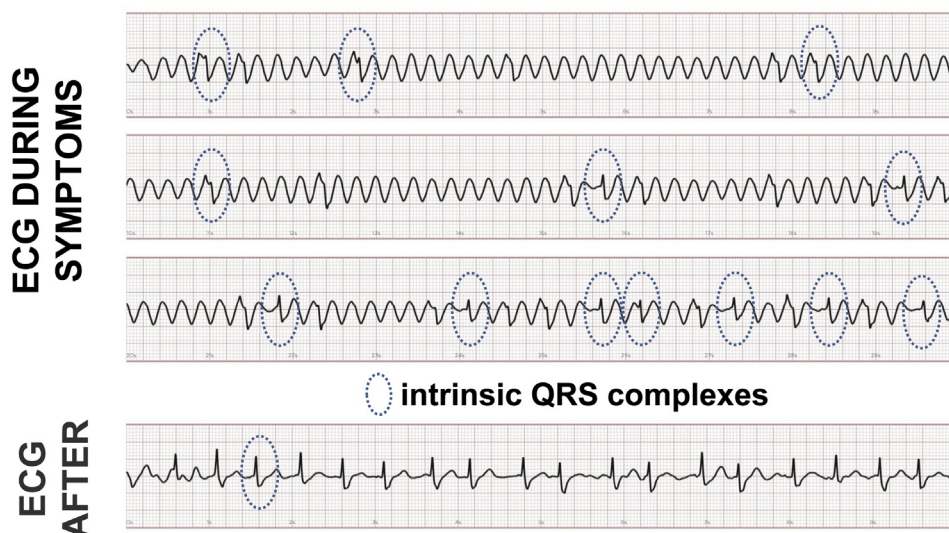
The incidence of dangerous rhythm disturbances is very low in the young and healthy, which form the majority of smartwatch wearers. A less than perfect screening test used in a population with a very low pretest probability of serious arrhythmias translates not only into a modest posttest probability of disease but also a real risk of anxiety, unnecessary diagnostic testing, hospitalization, and even potentially inappropriate treatments. Thus, every tracing consistent with sustained ventricular arrhythmia requires careful analysis for artifact. Recording a smartwatch ECG requires active patient participation for at least 30 seconds, and the patient is unlikely to remain conscious while in extremely fast ventricular tachycardia or ventricular fibrillation to accomplish this task.<sup>14</sup> On a 12-lead surface ECG, 3 findings should raise one's suspicion of artifact. First, the phenomenon is seen in only certain leads and not all leads. This first finding is unfortunately not applicable when interpreting single-lead smartwatch tracings. However, smartwatch ECGs are longer in duration and can be easily repeated, which may help eliminate potential sources of artifact. Second, the phenomenon is not compatible with a physiological process. For example, extremely fast and wide complexes are incompatible with physiological arrhythmias, particularly in the absence of severe symptoms. Third, the phenomenon does not disrupt

the underlying ventricular rate, meaning that the heart rate does not change in the presence and absence of the suspected artifact. Figure 5 shows an example of a smartwatch recording where artifacts could mislead a physician into a diagnosis of ventricular tachycardia or fibrillation. Intrinsic QRS complexes remain identifiable, however, confirming that the ventricular rate is actually at a normal rate.

Before any therapeutic decisions are made on the basis of watch recordings (eg, initiating anticoagulation for presumed AF or considering an implantable cardioverter-defibrillator for presumed ventricular tachycardia), it is imperative to ensure that the arrhythmia is real by carefully ruling out artifact or noise. Currently, published reports of wide QRS complex tachycardias recorded by smartwatches are rare.<sup>15</sup> Identifying the relationship between atrial and ventricular activity during a wide complex tachycardia can be among the most helpful feature to arrive at a diagnosis. During ventricular tachycardia, atrial activation is often independent of ventricular activation. In some cases, the atria are driven by the ventricles through retrograde conduction, but this can be difficult to appreciate on 12-lead ECGs, let alone a single-lead smartwatch ECGs, because of the small atrial deflections often being buried within the QRS complexes. When confronted with a smartwatch ECG with broad QRS complexes, it is important to interpret the tracing in the context of the patient's clinical presentation. Figure 6 shows a smartwatch ECG from a 61-year-old man with a history of anterior wall myocardial infarction with depressed left ventricular ejection fraction (40%) presenting with recurrent nonsyncopal episodes of palpitations. The patient had purchased a smartwatch and recorded a tracing during an episode, confirming the diagnosis of ventricular tachycardia.

Figure 7 shows a smartwatch ECG of a 19-year-old man hospitalized for syncope. He was able to record an ECG





**Figure 5** Smartwatch electrocardiogram (ECG) suggestive of wide QRS complex tachycardia. Careful inspection identifies intrinsic QRS complexes (dotted circles), confirming the presence of artifact.

with his smartwatch before losing consciousness. The tracing shows a characteristic pattern of AF conducted rapidly to the ventricles over an accessory pathway. A tachycardia is seen with short and irregular R-R intervals as well as variations in QRS morphology, ranging from relatively narrow to highly preexcited with significant cycle-to-cycle variations of the initial portion of the QRS (the delta wave), creating an ECG “accordion-like” pattern. This episode was treated with an external electric shock. A 12-lead ECG confirmed ventricular preexcitation, and he underwent radiofrequency ablation of his malignant accessory pathway.

The pretest probability of ventricular arrhythmia was relatively high for the first patient, and the tracing showed atrio-ventricular dissociation that was diagnostic of ventricular tachycardia. In the second case, the tracing shows characteristic features of AF in a patient with Wolff-Parkinson-White syndrome. It should be noted that this patient

had previously recorded a tracing in sinus rhythm that suggested an accessory pathway but he had not sent it to a doctor for interpretation. This raises the possibility of using smartwatches to prevent sudden death in young individuals. From a technical perspective, the smartwatch is capable of suggesting the diagnosis of certain channelopathies (eg, long QT), manifest accessory pathways, and certain structural heart diseases (eg, hypertrophic cardiomyopathy and arrhythmogenic right ventricular cardiomyopathy).<sup>16</sup>

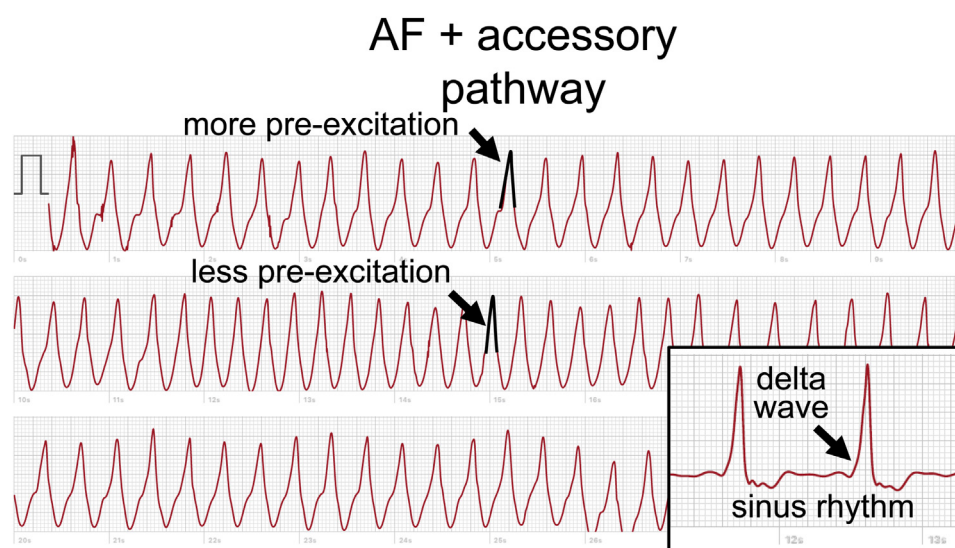
## G. Perspectives

Fundamental shifts in medicine are set to occur as the digital and connected era extends to the clinical realm. The Apple Watch is currently the dominant product with more than half of the global market share and more than 100 million watches sold. Since the launch of the Apple Watch Series

## ventricular tachycardia



**Figure 6** Regular, wide complex, monomorphic tachycardia at 110 beats/min. Dissociated and slower atrial activity (P waves identifiable on the tracing; arrows) can be appreciated, confirming the diagnosis of ventricular tachycardia.



**Figure 7** Atrial fibrillation with conduction over an accessory pathway. Irregular tachycardia with short and variable R-R intervals as well as variable QRS morphologies ranging from minimally to highly preexcited. AF = atrial fibrillation.

4, one of its objectives has clearly been to establish itself as the world leader in connected health, made clear by their current slogan “The future of health is on your wrist.” Currently their smartwatches enable activity measurements, sleep tracking, fall detection, heart rate sensors, blood oxygen sensors, and ECG recording. Upcoming features could include blood pressure measurements, blood sugar monitoring, diagnostics for sleep apnea, and body temperature monitoring (possibly through connected wearables/sensors). Their ECG technology could be extended by additional skin electrodes that communicate with the smartwatch to enable the recording of a 6- or 12-lead ECG. Fortunately, an increasing amount of competitors offer more affordable devices with similar features that are compatible with various smartphones. The combination of multiple sensors and big data could lead to artificial intelligence solutions where the smartwatch detects users at high risk of developing cardiovascular events. Integrating these new functionalities into the daily life of consumers raises important practical, logistical, and philosophical questions that remain unanswered. Widespread dissemination and use of these new devices likely requires buy-in by health professionals and probably a system of compensation for the time spent analyzing the digital data. Being able to handle larger volumes in the future is a 3-step process: (1) improving the accuracy of the proprietary algorithms and extending the automated diagnoses onboard the smartwatch, (2) implementing cloud-based automatic triage classification through a machine learning approach, and (3) appointing dedicated personnel (intramural or extramural) for postprocessing of the alerts and deciding whether clinical action is required. The medicolegal aspects of this process are similar to the current remote monitoring practice of implanted medical devices such as implantable loop recorders. The fact that the frequency of transmission is under the control of the patient may require a predefined

limitation of handled ECGs, but we have not yet experienced ECG overload from patients.

## Conclusion

Real-time access to ECGs, without a physician’s order, could allow individuals to play a more active role in their health care, facilitate screening efforts, and constitute a true revolution to our discipline. In the future, advances in artificial intelligence should reduce the number of false positives but could also lead to a significant expansion of automated diagnostic capabilities that are currently limited to differentiating between sinus rhythm and AF. Health care professionals should work with smartwatch manufacturers and critically examine the potential of this technology as its diagnostic boundaries are pushed.

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